

Project Proposal: Predicting Medical Insurance Charges

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Introduction

Medical insurance costs vary widely between people, and insurers and policyholders both benefit from understanding which personal factors drive those costs. This project aims to build a regression model that predicts a person's yearly insurance charges and identifies which factors, such as age, BMI, and smoking, have the biggest effect.

Dataset

I will use the Medical Cost Personal Dataset (Lantz, 2013), which has 1,338 rows of simulated US insurance beneficiaries; the target variable, charges, is continuous and ranges from about \$1,122 to \$63,770. The six predictors are three numeric variables (age, BMI, number of children) and three categorical variables (sex, smoker status, region).

Data cleaning

I will check for missing values and remove duplicate rows, convert the categorical variables into numeric dummy variables, and check numeric columns such as BMI for unrealistic outliers.

Initial visualizations

A histogram of charges will show whether the target is skewed, and scatter plots of charges against age and BMI (coloured by smoker status), along with box plots for each categorical variable, will show how each predictor relates to cost. A correlation heatmap will summarize the strength of these relationships and flag any predictors that are strongly correlated with each other.

Regression model

I will use multiple linear regression in Python (pandas, scikit-learn, statsmodels), because the target is continuous and there are several predictors whose individual effects can be interpreted from the coefficients. I will split the data into training and test sets and evaluate the model with R^2 and RMSE. Because charges are likely right-skewed, I will also test a log-transformed target and compare the results.

References

Lantz, B. (2013). *Machine learning with R*. Packt Publishing.

Stedy. (n.d.). *Machine-Learning-with-R-datasets* [Data set]. GitHub. Retrieved October 3, 2026, from <https://github.com/stedy/Machine-Learning-with-R-datasets>